Pharmacology SGD: Substance Abuse & Alcohols

1. A 36-year-old man who takes methadone 70 mg daily is seen after a motorcycle accident that fractured his femur. The orthopedic team starts a new analgesic. Within 2 hours he develops rhinorrhea, yawning, piloerection, abdominal cramping, and diarrhea. Pupils are 6 mm. Which of the following drugs was most likely started?

A.Oxycodone
B.Tramadol
C.Hydromorphone
D.Buprenorphine
E.Morphine

2. A 27-year-old man is brought to the emergency department after developing chest pressure at a party. Temperature is 38.2°C (100.8°F), pulse is 126/min, and blood pressure is 172/104 mm Hg. Pupils are 7 mm and reactive. He is diaphoretic and restless. ECG shows sinus tachycardia with ST depressions in the lateral leads. A friend says he inhaled a white powder through the nose that made his gums numb. Which of the following best describes the primary mechanism of the sympathomimetic effects of the substance most likely used?

A.Reversal of the vesicular monoamine transporter, causing cytoplasmic release of catecholamines
B.Inhibition of monoamine oxidase in presynaptic terminals
C.Direct agonism at postsynaptic alpha-1 receptors
D.Blockade of presynaptic reuptake transporters for norepinephrine, dopamine, and serotonin
E.Antagonism of presynaptic alpha-2 autoreceptors

3. A 49-year-old woman with chronic anxiety and insomnia, who takes alprazolam several times daily, is found minimally responsive at home. Respirations are 7/min and oxygen saturation is 86% on room air. The emergency physician gives an intravenous agent and her respiratory rate improves, but 10 minutes later she has a generalized tonic-clonic seizure. Which of the following best explains the seizure?

A.Competitive GABA-A receptor antagonism in a physically dependent patient
B.Direct activation of NMDA receptors by the administered agent
C.Inhibition of glycine receptors by the administered agent
D.Opioid receptor blockade leading to unopposed excitatory tone
E.Rebound hypoglycemia from the administered agent

4. A 22-year-old college student is brought in by campus police after being found shouting and swinging at bystanders. He is combative, has vertical and horizontal nystagmus, and does not appear to react to pain when restrained. Blood pressure is 168/98 mm Hg and pulse is 112/min. He cannot recall how he arrived at the party. Which of the following is the most likely mechanism of action of the drug responsible?

A.Partial agonism at the cannabinoid CB1 receptor
B.Noncompetitive antagonism at the NMDA receptor
C.Agonism at the 5-HT2A receptor
D.Positive allosteric modulation of GABA-A receptors
E.Agonism at the mu opioid receptor

5. A 56-year-old woman with alcohol dependence develops confusion, fever, tremors, and visual hallucinations 72 hours after stopping alcohol. She also has cirrhosis with impaired hepatic function. Which drug is most appropriate?

A. Diazepam
B. Chlordiazepoxide
C. Lorazepam
D. Disulfiram
E. Propranolol

6. A 38-year-old woman is treated with an oral antimicrobial for trichomoniasis. That evening she drinks two glasses of wine at dinner. Within 30 minutes she develops facial flushing, throbbing headache, nausea, and palpitations. Blood pressure is 98/60 mm Hg. Accumulation of which of the following is the most likely cause?

A.Acetate
B.Acetyl-CoA
C.Formaldehyde
D.Lactate
E.Acetaldehyde

7. A 44-year-old man presents with blurred vision, headache, and abdominal pain 12 hours after drinking an unlabeled liquid from a garage. Arterial pH is 7.14, anion gap is 28 mEq/L, and serum osmolal gap is elevated. Urinalysis shows no crystals. Fundoscopy shows optic disc hyperemia. Which of the following metabolites is most responsible for his visual findings?

A.Oxalic acid
B.Glycolic acid
C.Formic acid
D.Acetone
E.Acetaldehyde

8. A 46-year-old man who drinks heavily every day takes acetaminophen 1 g three times daily for 3 days for dental pain, which is within the labeled daily maximum. He presents with jaundice and right upper quadrant pain. AST is 4,800 U/L and ALT is 5,200 U/L. Which of the following best explains his susceptibility to hepatotoxicity at this dose?

A.Induction of CYP2E1, increasing production of a toxic metabolite
B.Inhibition of sulfotransferase by acetaldehyde
C.Increased renal reabsorption of acetaminophen
D.Impaired intestinal absorption leading to delayed clearance
E.Decreased hepatic blood flow from portal hypertension alone

9. A 47-year-old man who has been living in a shelter is brought to the emergency department with confusion. Examination shows bilateral lateral rectus weakness, wide-based gait, and disorientation to time. He has foul-smelling breath, and fingerstick glucose is 64 mg/dL. Which of the following is the most appropriate pharmacotherapy to administer first?

A.Naltrexone
B.Folic acid
C.Cyanocobalamin
D.Lorazepam
E.Thiamine

10. A 55-year-old man with alcohol use disorder wants help to remain abstinent after completing detoxification. He takes oxycodone daily for chronic back pain. Liver enzymes are normal and creatinine clearance is 90 mL/min. Which of the following is the most appropriate medication for relapse prevention?

A.Disulfiram
B.Naltrexone
C.Ondansetron
D.Buspirone
E.Acamprosate